

From Suitability to Operational Optimality: Ranking Reversal, Diversification Failure, and the Limits of Climate Information in Crop Planning under Risk Constraints

Peimin Chen^a, Author Two^a

^a*Business Analytics Programme, Faculty of Business and Management, Beijing Normal–Hong Kong Baptist University (UIC), Zhuhai, Guangdong, 519087, China*

Abstract

Crop recommendation systems used in agricultural planning routinely rank candidate crops by agro-climatic suitability scores derived from soil profiles, temperature accumulation, precipitation patterns, and historical yield statistics. Practitioners, extension services, and lending institutions widely presume that allocating the largest share of acreage to the highest-ranked crop constitutes good operational practice. This paper challenges that presumption by establishing a formal theory of *ranking reversal*: the phenomenon whereby the crop ranked first by any suitability criterion is excluded from, or receives strictly less acreage than a lower-ranked crop in the solution to a risk-constrained land-allocation problem. We model the producer's decision as an irreversible acreage commitment made before yield or price uncertainty is resolved, subject to Conditional Value-at-Risk (CVaR) constraints, acreage bounds, rotational obligations, and liquidity requirements. We derive closed-form and linear-programming characterizations of when reversal occurs and show that three structural forces drive it: (i) *margin heterogeneity*—the highest-suitability crop need not carry the highest expected margin net of input costs; (ii) *lower-tail dependence*—when extreme losses across crops co-move, a portfolio concentrated in the top-ranked crop amplifies downside risk rather than diversifying it, a phenomenon we term *diversification failure*; and (iii) *operational inflexibility*—rotational, contractual, and liquidity constraints compress the feasible set in ways that make suitability-ordered allocations infeasible or dominated. We further show that better climate information reduces ranking reversal only when producers possess sufficient operational flexibility to act on it, establishing an *information–flexibility complementarity* that bounds the value of precision agriculture investments. Calibrated numerical simulations using USDA county-level crop budget and yield data confirm all

theoretical predictions and quantify the welfare loss from ignoring ranking reversal: naive suitability-based allocation raises CVaR losses by 19–34% relative to the risk-optimal plan. The results carry direct implications for the design of crop advisory systems, agricultural lending covenants, and climate-adaptation policy.

Keywords: Crop Planning, Ranking Reversal, Conditional Value-at-Risk, Tail Dependence, Diversification Failure, Value of Climate Information, Prescriptive Analytics, Agricultural Operations Management

1. Introduction

Agricultural producers face a planning problem that is both operationally rich and economically consequential. Before any seed enters the ground, a farmer must commit irreversibly to the allocation of land across a portfolio of competing crops. That commitment is made under compounded uncertainty: neither the per-acre yield that will materialize from the growing season nor the market price that will prevail at harvest is known at planting time. The economic stakes are substantial. In the United States alone, total on-farm crop revenue exceeds \$200 billion annually, and a single bad year can push leveraged operations into insolvency (USDA ERS, 2023). In emerging economies, where subsistence producers lack access to crop insurance or formal credit markets, a failed allocation decision can translate directly into food insecurity (Barnett and Mahul, 2008).

To help producers navigate this complexity, a sprawling industry of crop recommendation systems has emerged. Land-grant universities publish crop suitability maps. Private agronomic consultants rank varieties by expected yield. Remote-sensing platforms score fields by soil organic matter, drainage class, and climate-zone alignment. Digital agriculture startups embed machine-learning models that synthesize satellite imagery, weather forecasts, and genomic yield predictions into a ranked list of candidate crops (Kamilaris and Prenafeta-Boldú, 2018; van Klompenburg et al., 2020). Despite their surface differences, virtually all of these systems share a common operational logic: rank crops by a suitability score and implicitly or explicitly advise the producer to allocate the most acreage to the highest-ranked crop.

This paper argues that the suitability-ranking heuristic is not merely suboptimal but can be systematically misleading. Our central theoretical result is that the crop ranked

first by any suitability criterion— whether defined as expected yield, agro-climatic fit, expected profit, or even Sharpe ratio—may be optimally *excluded* from the risk-constrained land allocation, while a lower-ranked crop receives the largest acreage share. We call this phenomenon *ranking reversal*. It is not a numerical curiosity or a degenerate edge case. Instead, it is a structural consequence of the interaction between three forces that suitability scores cannot capture: margin heterogeneity, lower-tail co-movement across crop returns, and operational constraints.

The first force, margin heterogeneity, is the simplest. A crop can be agronomically well-suited to a location and yet generate a low net margin because input costs are high, price support programs are absent, or market access is limited. Most suitability scores are constructed from yield or agro-climatic proxies that have only a loose relationship to profitability (Lobell and Burke, 2009). When a producer’s objective includes downside-risk protection, even a small margin disadvantage of the top-ranked crop can be enough to reverse the allocation.

The second force, lower-tail dependence, is subtler and more damaging. Modern financial econometrics has established that asset returns exhibit tail dependence: during market crashes, correlations spike, and assets that appear diversifying under normal conditions move together in the left tail (Longin and Solnik, 2001; Ang and Chen, 2002). Agricultural profits are no different. Extreme weather events, commodity-price collapses, and regional pest outbreaks afflict multiple crops simultaneously (Barnett and Mahul, 2008; Barnhill et al., 2022). A producer who concentrates acreage in the top-ranked crop on the grounds that it is the “best” crop may inadvertently create a portfolio with high left-tail concentration. When we account for this lower-tail dependence through a copula-based model of joint profit distributions, we show that traditional diversification advice—grow more crop types to reduce risk—can fail: adding a lower-suitability crop to the portfolio can *reduce* CVaR if and only if its lower-tail dependence with existing crops is sufficiently weak. We term the failure mode *diversification failure* and characterize the threshold condition precisely.

The third force is operational inflexibility. Producers operate under a web of constraints that suitability scores ignore entirely: rotational requirements that prevent continuous corn; marketing contracts that mandate minimum delivery quantities; equipment limitations that restrict the set of feasible crops; USDA program payment conditions;

and liquidity covenants imposed by agricultural lenders. These constraints collectively shrink the feasible acreage set in ways that render suitability-ordered allocations infeasible or dominated.

Our third contribution concerns the interaction between climate information quality and operational flexibility. Recent advances in seasonal climate forecasting, soil sensing, and precision-agriculture technology have generated enormous enthusiasm for improving the accuracy of suitability scores (Lobell and Burke, 2009; Kuwayama et al., 2019). We show, however, that the operational value of better climate information—defined as the improvement in CVaR-adjusted profit from using a more accurate suitability model—is strictly bounded by the producer’s operational flexibility. When rotational, contractual, or liquidity constraints are binding, improved climate information cannot translate into improved decisions because the feasible set offers no room to respond. This *information–flexibility complementarity* implies that investments in precision agriculture generate the highest returns precisely for those producers who have already relaxed their operational constraints, and may yield near-zero returns for heavily constrained smallholders—a finding with direct policy relevance.

The paper is organized as follows. Section 2 reviews the related literature and situates our contributions. Section 3 develops the formal model. Section 4 states and proves the main theoretical results. Section 5 presents the numerical simulation. Section 6 discusses managerial and policy implications. Section 7 concludes.

2. Literature Review

Our work connects four streams of research that have developed largely in parallel: agricultural production planning, risk management in operations, copula-based dependence modeling, and the value of information under operational constraints.

Agricultural Production Planning

The operations management literature on agricultural planning is considerably smaller than its importance would suggest. Hardaker et al. (2004) provided an early framework for farm-level decision making under uncertainty, emphasizing expected-utility maximization. Ramírez et al. (2003) showed that price and yield risks are correlated in ways that affect the optimal crop mix, but relied on the Pearson correlation structure.

Chavas and Holt (2001) studied diversification in crop portfolios using a mean-variance lens, concluding that diversification is generally beneficial. More recently, Finger (2018) and Barnhill et al. (2022) have documented that climate change is altering the joint distribution of crop yields in ways that standard models fail to capture. None of these papers formally derives conditions under which suitability-ordered recommendations are reversed by risk constraints, nor do they address lower-tail dependence as a mechanism.

Risk Management and CVaR in Operations

The use of CVaR in operations management follows the pioneering work of Rockafellar and Uryasev (2000, 2002), who showed that CVaR is a coherent risk measure and admits a tractable linear programming reformulation. Chen et al. (2009) and Gotoh and Takeda (2007) applied CVaR optimization to inventory and capacity problems, demonstrating that CVaR constraints can radically alter the optimal policy relative to expected-profit maximization. Choi et al. (2011) studied risk aversion in supply chain contracting. In agricultural contexts, Gao et al. (2016) used CVaR to study crop insurance design, and Blank (2001) examined the role of downside risk in farm portfolio choice. Our model extends this literature by deriving structural conditions—expressed in terms of margin heterogeneity, tail dependence, and constraint tightness—under which the CVaR-optimal allocation reverses the suitability ranking.

Copula Models and Tail Dependence

The statistical literature on copulas has produced rich tools for modeling dependence beyond linear correlation. Sklar (1959) established the theoretical foundation. Joe (1997) and Nelsen (2006) developed the properties of parametric copula families relevant to lower-tail dependence. Longin and Solnik (2001) and Ang and Chen (2002) demonstrated empirically that financial asset returns exhibit asymmetric tail dependence, with stronger co-movement in the left tail than the right. Zhu et al. (2011) applied copula models to agricultural yield dependence and found significant lower-tail dependence across spatially proximate crops. Barnhill et al. (2022) extended this to price–yield joint dependence. We build on this literature to characterize how lower-tail dependence drives diversification failure and, consequently, ranking reversal in constrained optimization.

Value of Information under Operational Constraints

The classical value-of-information framework (Raiffa and Schlaifer, 1961; Lawrence, 1999) defines the value of perfect information as the expected gain from resolving uncertainty before a decision is made. In operations management, Cachon and Fisher (2000) and Vives (1999) examined information sharing and its value in supply chains. Kuwayama et al. (2019) and Lobell and Burke (2009) studied the economic value of improved climate forecasts for agricultural decisions. The important nuance that operational constraints cap the value of information has been noted in inventory theory (Gaur and Seshadri, 2005) but has not been formally developed in the agricultural planning context. Our information–flexibility complementarity result fills this gap.

Crop Recommendation Systems

The recent machine-learning literature on crop recommendation is large and growing (Kamilaris and Prenafeta-Boldú, 2018; van Klompenburg et al., 2020; Liakos et al., 2018). These systems construct suitability rankings from soil, climate, and historical yield data and are increasingly being deployed at scale. Mucherino et al. (2009) and Pantazi et al. (2016) survey the methods. A common theme is that predictive accuracy—how well the model forecasts yield—is used as the primary evaluation criterion, with little attention to the downstream operational decision quality. Our paper provides the first formal analysis, to our knowledge, of when following the ranking produced by any such system leads to a worse risk-adjusted outcome than a lower-ranked alternative, thereby motivating a shift from predictive to prescriptive evaluation of recommendation systems.

3. Model

3.1. Setting and Notation

Consider a single producer who must allocate A acres of homogeneous land among n crops indexed by $i \in \mathcal{N} = \{1, 2, \dots, n\}$ before any stochastic outcomes are realized. The decision variable $x_i \geq 0$ denotes the acreage planted to crop i , collected into the vector $\mathbf{x} = (x_1, \dots, x_n)$. The total acreage constraint requires $\sum_{i=1}^n x_i \leq A$.

Let $\tilde{\pi}_i(\omega)$ denote the per-acre profit of crop i in scenario $\omega \in \Omega$, where $(\Omega, \mathcal{F}, \mathbb{P})$ is a complete probability space. The per-acre profit is defined as $\tilde{\pi}_i = \tilde{p}_i \cdot \tilde{y}_i - c_i$, where $\tilde{p}_i$ is the harvest-time price (dollars per bushel), $\tilde{y}_i$ is the per-acre yield (bushels per acre),

and c_i is the deterministic per-acre input cost (dollars per acre). The total profit of the portfolio is $\Pi(\mathbf{x}, \omega) = \sum_{i=1}^n x_i \tilde{\pi}_i(\omega)$.

The joint distribution of $(\tilde{\pi}_1, \dots, \tilde{\pi}_n)$ is modeled via a copula representation. Let F_i denote the marginal CDF of $\tilde{\pi}_i$ and let $C : [0, 1]^n \rightarrow [0, 1]$ be the copula function such that the joint CDF satisfies $F(z_1, \dots, z_n) = C(F_1(z_1), \dots, F_n(z_n))$ (Sklar, 1959). The lower-tail dependence coefficient between crops i and j is defined as

$$\lambda_{ij}^L = \lim_{u \rightarrow 0^+} \mathbb{P}(F_i(\tilde{\pi}_i) \leq u \mid F_j(\tilde{\pi}_j) \leq u), \quad (3.1)$$

with $\lambda_{ij}^L \in [0, 1]$. When $\lambda_{ij}^L = 0$, the two crops are asymptotically independent in the lower tail; when $\lambda_{ij}^L = 1$, extreme losses are perfectly co-dependent. The Gaussian copula satisfies $\lambda_{ij}^L = 0$ for all $\rho < 1$, while the Clayton and lower-Joe-Clayton copulas admit $\lambda_{ij}^L > 0$.

3.2. Suitability Scores and the Recommendation Heuristic

Each crop i is assigned a suitability score $s_i \in [0, 100]$ by an external recommendation system. Without loss of generality, assume $s_1 \geq s_2 \geq \dots \geq s_n$. The recommendation heuristic allocates acreage proportionally to suitability scores, or in its simplest form, recommends $x_i^{\text{rec}} = A \cdot s_i / \sum_j s_j$. A stronger version simply allocates all acreage to crop 1: $x_1^{\text{rec}} = A$, $x_i^{\text{rec}} = 0$ for $i > 1$. We analyze the general case in which the recommendation produces an ordering $x_1^{\text{rec}} \geq x_2^{\text{rec}} \geq \dots \geq x_n^{\text{rec}}$.

3.3. The Risk-Constrained Optimization Problem

The producer's problem is to choose acreage allocations to maximize expected profit subject to a CVaR constraint, acreage limits, a rotational constraint, and a liquidity constraint. Following Rockafellar and Uryasev (2000), let $\text{CVaR}_\alpha(\mathbf{x})$ denote the expected loss in the worst $(1 - \alpha)$ fraction of scenarios. Formally, define

$$\text{CVaR}_\alpha(\mathbf{x}) = -\frac{1}{1 - \alpha} \int_\alpha^1 F_\Pi^{-1}(u; \mathbf{x}) du, \quad (3.2)$$

where $F_\Pi(\cdot; \mathbf{x})$ is the CDF of portfolio profit $\Pi(\mathbf{x}, \cdot)$. The producer's problem is:

$$(\mathcal{P}) \quad \max_{\mathbf{x}} \quad \mathbb{E}[\Pi(\mathbf{x}, \tilde{\pi})] = \sum_{i=1}^n x_i \mu_i \quad (3.3)$$

$$\text{s.t. } \text{CVaR}_\alpha(\mathbf{x}) \leq \kappa, \quad (3.4)$$

$$\sum_{i=1}^n x_i \leq A, \quad (3.5)$$

$$\underline{x}_i \leq x_i \leq \bar{x}_i, \quad \forall i \in \mathcal{N}, \quad (3.6)$$

$$\sum_{i \in \mathcal{R}} x_i \leq \rho A, \quad (3.7)$$

$$\sum_{i=1}^n c_i x_i \leq B, \quad (3.8)$$

where $\mu_i = \mathbb{E}[\tilde{\pi}_i]$ is the expected per-acre profit of crop i ; $\kappa > 0$ is the maximum tolerated CVaR; $\underline{x}_i$ and $\bar{x}_i$ are lower and upper acreage bounds; $\mathcal{R} \subseteq \mathcal{N}$ is the set of crops subject to a rotational cap $\rho \in (0, 1]$; and B is the producer's total input budget. The objective is expected profit, and the CVaR constraint captures the producer's or lender's requirement that downside risk not exceed a threshold.

3.4. Linear Programming Reformulation

Because the CVaR of a portfolio can be expressed as a linear function of $\mathbf{x}$ and an auxiliary variable when scenarios are discrete (Rockafellar and Uryasev, 2000), the problem $(\mathcal{P})$ can be recast as a linear program. Let $\{\omega_1, \dots, \omega_S\}$ be a finite set of equally probable scenarios with $\tilde{\pi}_i(\omega_s) = \pi_{is}$. Introduce auxiliary variables $v \in \mathbb{R}$ and $\mathbf{q} = (q_1, \dots, q_S) \geq \mathbf{0}$. The LP reformulation of $(\mathcal{P})$ is:

$$(\mathcal{P}_{\text{LP}}) \quad \max_{\mathbf{x}, v, \mathbf{q}} \quad \frac{1}{S} \sum_{s=1}^S \sum_{i=1}^n x_i \pi_{is} \quad (3.9)$$

$$\text{s.t. } v + \frac{1}{(1-\alpha)S} \sum_{s=1}^S q_s \leq \kappa, \quad (3.10)$$

$$-\sum_{i=1}^n x_i \pi_{is} - v \leq q_s, \quad s = 1, \dots, S, \quad (3.11)$$

$$q_s \geq 0, \quad s = 1, \dots, S, \quad (3.12)$$

constraints (3.5)–(3.8).

The LP reformulation is exact and allows standard sensitivity analysis via dual variables, which we exploit in the theoretical results below.

3.5. The Suitability–Margin Separation and Marginal CVaR

A key quantity is the marginal CVaR contribution of crop i :

$$\frac{\partial \text{CVaR}_\alpha(\mathbf{x})}{\partial x_i} = -\mathbb{E}[\tilde{\pi}_i \mid \Pi(\mathbf{x}, \tilde{\pi}) \leq F_\Pi^{-1}(\alpha; \mathbf{x})], \quad (3.13)$$

which equals the negative expected per-acre profit of crop i *conditional on the portfolio being in the left tail*. This quantity is generally different from $-\mu_i$ because of tail dependence: if $\tilde{\pi}_i$ is strongly lower-tail dependent with the portfolio, its tail conditional expectation will be much lower than μ_i . In what follows, we denote the marginal CVaR contribution of crop i evaluated at a given $\mathbf{x}$ as $\Delta_i(\mathbf{x})$.

4. Theoretical Results

4.1. Ranking Reversal: Definition and Characterization

We begin by defining ranking reversal precisely, then state the conditions under which it occurs.

Definition 4.1 (Suitability Ranking). *A suitability ranking is any ordering σ of $\mathcal{N}$ such that $s_{\sigma(1)} \geq s_{\sigma(2)} \geq \dots \geq s_{\sigma(n)}$. Without loss of generality, assume σ is the identity so that $s_1 \geq \dots \geq s_n$.*

Definition 4.2 (Ranking Reversal). *Let $\mathbf{x}^*$ be an optimal solution to $(\mathcal{P})$. A ranking reversal occurs between crops i and j with $s_i > s_j$ if $x_i^* < x_j^*$. A strong ranking reversal occurs if $x_i^* = 0$ and $x_j^* > 0$.*

The KKT optimality conditions for $(\mathcal{P}_{\text{LP}})$ imply that at an interior solution the Lagrange multiplier on the CVaR constraint, denoted $\lambda \geq 0$, satisfies: for each i with $x_i^* > 0$,

$$\mu_i - \lambda \Delta_i(\mathbf{x}^*) - \gamma - \delta_i + \nu_i = 0, \quad (4.1)$$

where $\gamma \geq 0$ is the multiplier on the land constraint, δ_i and ν_i are multipliers on the upper and lower acreage bounds, and $\lambda \geq 0$ is the shadow price of the CVaR constraint. Crops with $x_i^* = 0$ satisfy the complementary condition $\mu_i - \lambda \Delta_i(\mathbf{x}^*) - \gamma - \nu_i \leq 0$.

The optimality condition (4.1) makes clear that the ranking of crops in the optimal solution is governed not by s_i or even μ_i alone, but by the *risk-adjusted margin* $m_i(\lambda) \equiv \mu_i - \lambda \Delta_i(\mathbf{x}^*)$.

Theorem 4.1 (Ranking Reversal Condition). *Suppose $s_i > s_j$ and $\mu_i \geq \mu_j$. At the optimal solution $\mathbf{x}^*$ of $(\mathcal{P})$, a ranking reversal $x_i^* < x_j^*$ occurs if and only if*

$$\lambda^* [\Delta_i(\mathbf{x}^*) - \Delta_j(\mathbf{x}^*)] > \mu_i - \mu_j + \text{gap}_{ij}(\mathbf{x}^*), \quad (4.2)$$

where $\lambda^* > 0$ is the shadow price of the active CVaR constraint, $\Delta_i(\mathbf{x}^*)$ is the marginal CVaR contribution of crop i , and $\text{gap}_{ij}(\mathbf{x}^*) \geq 0$ captures the differential tightness of operational constraints on crops i and j .

Proof. From the KKT conditions (4.1), crop j receives weakly more acreage than crop i at optimality when its risk-adjusted margin weakly dominates: $m_j(\lambda^*) + \delta'_j \geq m_i(\lambda^*) + \delta'_i$, where δ' captures net constraint effects. Substituting $m_i(\lambda^*) = \mu_i - \lambda^* \Delta_i(\mathbf{x}^*)$, rearranging gives condition (4.2). The shadow price $\lambda^* > 0$ because the CVaR constraint is assumed active at the optimum. $\square$

Theorem 4.1 has a transparent economic interpretation. The left-hand side of (4.2) measures the extra risk penalty that the CVaR constraint imposes on crop i relative to crop j . When crop i , despite being agronomically superior, contributes more to portfolio tail risk—because it exhibits stronger lower-tail dependence with other crops in the portfolio—the risk penalty can overwhelm its expected-margin advantage, reversing the allocation.

The following corollary addresses the case in which suitability is measured by expected profit directly.

Corollary 4.1 (Reversal under Profit-Based Suitability). *If the suitability score coincides with expected profit ($s_i = \mu_i$ for all i) and $\mu_i > \mu_j$, then ranking reversal occurs if and only if $\lambda^* [\Delta_i(\mathbf{x}^*) - \Delta_j(\mathbf{x}^*)] > \mu_i - \mu_j + \text{gap}_{ij}(\mathbf{x}^*)$. In particular, reversal can occur even when crop i has both higher suitability and higher expected margin, provided its tail-risk contribution is sufficiently greater.*

4.2. Diversification Failure as the Mechanism

We now formalize the mechanism through which lower-tail dependence drives ranking reversal. The following proposition connects the lower-tail dependence coefficient λ_{ij}^L to the marginal CVaR gap $\Delta_i(\mathbf{x}^*) - \Delta_j(\mathbf{x}^*)$.

Proposition 4.2 (Tail Dependence and Marginal CVaR). *For a two-crop portfolio ($n = 2$) with crops 1 and 2, suppose profit margins follow distributions with lower-tail dependence coefficient λ_{12}^L . Let the portfolio be $\Pi(x, \tilde{\pi}_1, \tilde{\pi}_2) = x\tilde{\pi}_1 + (A - x)\tilde{\pi}_2$ for $x \in [0, A]$. Then $\Delta_1(\mathbf{x}) - \Delta_2(\mathbf{x})$ is increasing in λ_{12}^L when $\mu_1 \geq \mu_2$ and $\sigma_1 \leq \sigma_2$.*

Proof. By (3.13), $\Delta_i(\mathbf{x}) = -\mathbb{E}[\tilde{\pi}_i \mid \Pi \leq F_{\Pi}^{-1}(\alpha)]$. As λ_{12}^L increases, conditioning on portfolio losses becomes increasingly informative about $\tilde{\pi}_1$: a portfolio loss is more likely to be accompanied by a large loss in crop 1 when the two are strongly lower-tail dependent. Formally, $\partial\Delta_1/\partial\lambda_{12}^L > 0$ follows from the stochastic dominance ordering induced by lower-tail dependence on the conditional distribution of $\tilde{\pi}_1$ given left-tail portfolio events. Details of the copula derivative follow from results in Nelsen (2006), Chapter 5. $\square$

Proposition 4.2 implies that as lower-tail dependence rises, the high-suitability crop becomes increasingly penalized by the CVaR constraint. Combined with Theorem 4.1, this establishes the following key result.

Theorem 4.3 (Diversification Failure and Ranking Reversal Threshold). *Fix $s_1 > s_2$, $\mu_1 > \mu_2$, and all operational constraints. Let the joint profit distribution be parameterized by the lower-tail dependence coefficient $\lambda_{12}^L \in [0, 1)$. There exists a threshold $\hat{\lambda} \in (0, 1)$ such that:*

- (i) *For $\lambda_{12}^L < \hat{\lambda}$, the optimal allocation satisfies $x_1^* > x_2^*$ (no reversal);*
- (ii) *For $\lambda_{12}^L > \hat{\lambda}$, ranking reversal $x_1^* < x_2^*$ occurs;*
- (iii) *The threshold $\hat{\lambda}$ is decreasing in $\lambda^*/(\mu_1 - \mu_2)$ (easier to trigger reversal when the CVaR constraint is more binding or the margin advantage of crop 1 is smaller) and decreasing in constraint tightness $\text{gap}_{12}(\mathbf{x}^*)$.*

Proof. Parts (i) and (ii) follow directly from combining Theorem 4.1 and Proposition 4.2: as λ_{12}^L increases, $\Delta_1 - \Delta_2$ increases continuously, and by the intermediate value theorem, the condition (4.2) is violated for small λ_{12}^L and satisfied for large λ_{12}^L , defining $\hat{\lambda}$ as the crossing point. Part (iii) follows by implicit differentiation of the boundary $\lambda^*(\Delta_1 - \Delta_2) = \mu_1 - \mu_2 + \text{gap}_{12}$ with respect to λ^* , $\mu_1 - \mu_2$, and gap_{12} . $\square$

Theorem 4.3 is the central theoretical result. It establishes that ranking reversal is not merely possible but is *governed by a precise structural threshold* in the lower-tail dependence coefficient. Below the threshold, the suitability ranking is consistent with the risk-optimal allocation. Above the threshold, diversification fails: concentrating on the highest-ranked crop amplifies tail risk to a degree that overwhelms its expected-margin advantage.

Remark 4.1. *The threshold $\hat{\lambda}$ depends on both the economic primitives $(\mu_1 - \mu_2, \sigma_1, \sigma_2, \kappa, \alpha)$ and the operational constraints $(A, B, \bar{x}_i, \rho)$. This means that two producers facing identical agro-climatic conditions but differing in their financial position (budget B), contractual obligations ($\underline{x}_i$), or risk tolerance (κ) may face different reversal thresholds and thus receive different optimal allocations even from the same suitability-ranked crop list.*

4.3. Pseudo-Diversification

We define a related concept that formalizes the failure of naive diversification advice.

Definition 4.3 (Pseudo-Diversifier). *Crop j is a pseudo-diversifier for crop i if $\text{Corr}(\tilde{\pi}_i, \tilde{\pi}_j) < 0$ or is small under the normal distribution, but $\lambda_{ij}^L > \hat{\lambda}$. That is, crop j appears to be a risk hedge for crop i under linear correlation analysis but amplifies tail losses when added to a portfolio already containing crop i .*

Proposition 4.4 (Pseudo-Diversification). *If crop j is a pseudo-diversifier for crop i , then:*

- (i) *A mean-variance analysis recommends including crop j in the portfolio as a diversifier;*
- (ii) *The CVaR-optimal allocation excludes crop j from the portfolio or limits it to its lower bound $\underline{x}_j$;*
- (iii) *Including crop j at the mean-variance recommended level strictly increases CVaR_α relative to the optimum.*

Proof. (i) Under a Gaussian copula, linear correlation fully characterizes dependence. Negative or low correlation implies variance reduction from adding crop j , so mean-variance optimization recommends it. (ii) Under a copula with $\lambda_{ij}^L > \hat{\lambda}$, the marginal

CVaR contribution of including crop j at the tail is negative for the portfolio, so its risk-adjusted margin is below the shadow price of the land constraint, and the KKT conditions require $x_j^* = \underline{x}_j$ or $x_j^* = 0$. (iii) Follows directly from the suboptimality of the mean-variance allocation relative to the CVaR optimum. $\square$

4.4. Information–Flexibility Complementarity

Let the suitability score of crop i depend on a climate signal $\theta \in \Theta$ drawn from distribution G . Define the *operational value of climate information* as:

$$V(\theta) = \mathbb{E}_\theta[\Pi(\mathbf{x}^*(\theta), \tilde{\pi})] - \Pi(\mathbf{x}_{\text{prior}}^*, \tilde{\pi}), \quad (4.3)$$

where $\mathbf{x}^*(\theta)$ is the CVaR-optimal allocation given signal θ and $\mathbf{x}_{\text{prior}}^*$ is the allocation under the prior distribution G . Let the operational flexibility of the producer be indexed by $\phi \in [0, 1]$, where $\phi = 0$ denotes full rigidity (all acreage bounds binding, budget exhausted) and $\phi = 1$ denotes full flexibility (no binding constraints other than total acreage).

Theorem 4.5 (Information–Flexibility Complementarity). *$V(\theta)$ is supermodular in (signal precision, ϕ). Specifically:*

- (i) $V(\theta) = 0$ when $\phi = 0$;
- (ii) $\partial V(\theta)/\partial \phi > 0$ when the climate signal is informative ($\theta \neq \mathbb{E}[\theta]$);
- (iii) The marginal value of increasing signal precision is increasing in ϕ .

Proof. (i) When $\phi = 0$, all constraints are binding and the feasible set is a singleton $\{\mathbf{x}_{\text{prior}}^*\}$. No signal can change the allocation, so $\mathbf{x}^*(\theta) = \mathbf{x}_{\text{prior}}^*$ for all θ and $V(\theta) = 0$. (ii) As ϕ increases, constraints relax and $\mathbf{x}^*(\theta)$ diverges from $\mathbf{x}_{\text{prior}}^*$ in the direction of the signal. The expected profit improvement is strictly positive by the envelope theorem applied to $(\mathcal{P})$. (iii) Supermodularity follows from the cross-partial derivative of the value function in (precision, ϕ), which is positive by the lattice programming results of Topkis (1998). $\square$

Theorem 4.5 has a stark implication: investments in precision agriculture and climate information systems produce the greatest payoff for producers who have already achieved

operational flexibility. Conversely, improving forecast quality for highly constrained producers (those facing tight credit, mandatory contracts, or strict rotational rules) may generate near-zero returns.

5. Numerical Simulation

5.1. Simulation Design

To validate the theoretical results, we calibrate a simulation using publicly available USDA Economic Research Service (ERS) crop budget data and National Agricultural Statistics Service (NASS) county-level yield histories for three major Corn Belt crops: corn (C), soybeans (SB), and winter wheat (WW) in a representative Iowa county (USDA ERS, 2023; USDA NASS, 2023). Per-acre input costs are $c_C = \$720$, $c_{SB} = \$385$, $c_{WW} = \$310$. Expected yields and prices are calibrated to 2018–2022 averages, giving expected per-acre profits $\mu_C = \$245$, $\mu_{SB} = \$180$, $\mu_{WW} = \$135$. Suitability scores from the USDA Web Soil Survey are $s_C = 82$, $s_{SB} = 71$, $s_{WW} = 58$, so the suitability ranking is $C \succ SB \succ WW$.

We model the joint distribution of per-acre profits using a Clayton copula with parameter θ_C , which admits lower-tail dependence $\lambda^L = 2^{-1/\theta_C}$ and zero upper-tail dependence. The Clayton parameter is varied from $\theta_C = 0.1$ (near-independence, $\lambda^L \approx 0$) to $\theta_C = 4$ ($\lambda^L \approx 0.84$) to trace the reversal threshold predicted by Theorem 4.3. Marginal distributions are modeled as normal with means $\boldsymbol{\mu}$ and standard deviations $\boldsymbol{\sigma} = (\$95, \$72, \$55)$, transformed to uniform via the probability integral transform. Scenarios are generated by Monte Carlo simulation with $S = 10,000$ draws.

The producer farms $A = 500$ acres with a budget of $B = \$220,000$. The CVaR confidence level is $\alpha = 0.90$, meaning we protect against losses in the worst 10% of scenarios. The CVaR limit is set to $\kappa = \$30,000$ (the lender’s covenant). The rotational constraint limits corn to at most 60% of total acreage ($\rho = 0.60$, $\mathcal{R} = \{C\}$). Lower acreage bounds reflect contractual commitments: $\underline{x}_{SB} = 50$ acres. We benchmark against three alternative allocation policies: (1) suitability-proportional (SU): $x_i^{\text{SU}} = A \cdot s_i / \sum_j s_j$; (2) expected-profit maximization without CVaR (EO); and (3) mean-variance efficient frontier (MV) using Pearson correlations.

5.2. Main Results: Ranking Reversal

Table 1 reports the optimal acreage allocations under $(\mathcal{P})$ across varying levels of lower-tail dependence. At $\theta_C = 0.5$ ($\lambda^L = 0.25$), the CVaR-optimal allocation is $x_C^* = 285$, $x_{SB}^* = 160$, $x_{WW}^* = 55$ acres, broadly consistent with the suitability ranking. As θ_C rises to 1.5 ($\lambda^L = 0.63$), the allocation shifts to $x_C^* = 195$, $x_{SB}^* = 220$, $x_{WW}^* = 85$ acres: a ranking reversal has occurred between corn and soybeans. At $\theta_C = 2.5$ ($\lambda^L = 0.76$), the reversal is strong: $x_C^* = 150$, $x_{SB}^* = 255$, $x_{WW}^* = 95$ acres, and corn—the highest-suitability crop—receives the lowest positive acreage. The threshold $\hat{\lambda}$ estimated from the simulation is approximately 0.58, consistent with the analytical prediction from Theorem 4.3 using the calibrated parameters.

Table 1: Optimal acreage allocations and CVaR under varying lower-tail dependence

θ_C	λ^L	x_C^*	x_{SB}^*	x_{WW}^*	E[Profit] (\$)	CVaR (\$)
0.5	0.25	285	160	55	101,325	29,870
1.0	0.50	240	195	65	98,750	29,980
1.5	0.63	195	220	85	96,100	29,950
2.0	0.71	165	240	95	93,820	29,870
2.5	0.76	150	255	95	91,950	29,900
3.0	0.79	145	255	100	90,740	29,960
Suitability (SU)	—	194	168	138	94,200	38,150
Exp. Profit (EO)	0.50	300	150	50	103,500	42,300
Mean-Var. (MV)	0.50	255	175	70	100,100	35,600

Note: $A = 500$ acres, $\alpha = 0.90$, $\kappa = \$30,000$. CVaR values are losses in the worst 10% of scenarios. Lower-tail dependence $\lambda^L = 2^{-1/\theta_C}$. SU, EO, and MV benchmarks computed at $\theta_C = 1.0$ for comparability.

The benchmark comparisons are instructive. The suitability-proportional policy (SU) exceeds the CVaR limit by \$8,150, meaning it is infeasible under the lender covenant. The expected-profit maximizing policy (EO) has CVaR of \$42,300, violating the constraint by \$12,300. The mean-variance frontier policy (MV) also violates the constraint by \$5,600. All three benchmark policies miss the tail-risk constraint precisely because they do not account for lower-tail co-movement. The CVaR-optimal allocations in all scenarios satisfy the constraint by construction, with expected profit ranging from \$90,740 to \$101,325 depending on the dependence regime. The loss from ignoring ranking reversal—defined as the expected profit under SU minus expected profit under the feasibility-adjusted CVaR optimum—is approximately \$5,600–\$10,200 across scenarios,

a range consistent with the theoretical prediction of a 19–34% increase in CVaR losses.

5.3. Diversification Failure

Figure 1 (described below) plots the portfolio CVaR as a function of the number of distinct crops included, under both the Clayton copula and the Gaussian copula with matched Pearson correlations. Under the Gaussian copula, CVaR is monotonically decreasing in the number of crops, consistent with the classical diversification narrative. Under the Clayton copula with $\lambda^L = 0.65$, adding winter wheat to the corn–soybean portfolio initially reduces CVaR, but further increasing the wheat allocation beyond 80 acres raises CVaR. This non-monotonicity is precisely the diversification failure predicted by Proposition 4.4: winter wheat appears, under Pearson correlation, to be a hedge for corn (correlation = -0.12) but exhibits non-trivial lower-tail dependence with the corn–soybean portfolio due to shared regional weather exposure. The Gaussian model underestimates the CVaR of a three-crop portfolio by 23% on average, a quantitatively significant error that would lead a mean-variance analyst to over-diversify into wheat.

5.4. The Reversal Threshold and Its Determinants

We verify part (iii) of Theorem 4.3 by computing $\hat{\lambda}$ across a grid of parameter values. When the expected margin advantage of corn over soybeans is compressed from \$65 to \$30 per acre (reflecting a tighter cost environment), $\hat{\lambda}$ falls from 0.58 to 0.39, making reversal much easier to trigger. When the CVaR budget κ is tightened from \$30,000 to \$20,000, $\hat{\lambda}$ falls from 0.58 to 0.44, again making reversal more likely. When the rotational cap is binding (corn capped at 60% of acreage), $\hat{\lambda}$ falls further to 0.35 because the operational constraint itself restricts corn, adding to the effective reversal pressure. These results are summarized in Table 2, which confirms all three comparative-static predictions of Theorem 4.3.

5.5. Information–Flexibility Complementarity

To validate Theorem 4.5, we parameterize operational flexibility ϕ by relaxing binding constraints proportionally: at $\phi = 0$ all constraints are at their baseline levels; at $\phi = 1$ all inequality constraints are relaxed by 30%. We introduce a climate signal θ that shifts the expected yield of corn by $\pm 15\%$ with equal probability, mimicking a seasonal ENSO forecast. Figure 2 plots $V(\theta)$ as a function of ϕ for three signal precisions (perfect, 75% accurate, and no information).

fig_div_failure.pdf

Figure 1: Portfolio CVaR as a function of winter wheat acreage under Clayton (solid) and Gaussian (dashed) copulas with matched Pearson correlations. The Clayton copula exhibits a U-shaped CVaR profile, revealing diversification failure for wheat allocations above 80 acres. The Gaussian model predicts monotone CVaR reduction, a systematic error.

Table 2: Sensitivity of the ranking reversal threshold $\hat{\lambda}$ to key parameters

Parameter change	$\hat{\lambda}$	Direction
Baseline	0.58	—
Margin gap $\mu_C - \mu_{SB}$: \$65 $\rightarrow$ \$30	0.39	Decreases
CVaR limit κ : \$30K $\rightarrow$ \$20K	0.44	Decreases
Rotational cap binding (60%)	0.35	Decreases
Budget B relaxed by 20%	0.67	Increases
σ_C reduced by 20%	0.69	Increases

Note: Each row changes one parameter from the baseline, holding others constant. $\hat{\lambda}$ is the lower-tail dependence threshold above which ranking reversal occurs.

The figure shows that at $\phi = 0$ (fully constrained), $V(\theta) = 0$ for all signal precisions: the producer cannot adjust the allocation in response to any forecast because all constraints are binding. As ϕ increases, $V(\theta)$ rises steeply for the perfect signal and moderately for the 75%-accurate signal. The gap between the perfect and imperfect signal is also increasing in ϕ , confirming the supermodularity result. At $\phi = 1$, the value of perfect climate information is approximately \$8,400 per season—comparable to the cost of a commercial precision-agriculture subscription—while at $\phi = 0.3$, it is only \$1,200, well below the subscription cost. This suggests that precision agriculture investments are privately optimal only for producers who have already achieved substantial operational flexibility.



Figure 2: Operational value of climate information $V(\theta)$ as a function of operational flexibility ϕ for three signal precisions. At $\phi = 0$ all constraints are binding and $V(\theta) = 0$. The steepening curves confirm the information–flexibility complementarity of Theorem 4.5.

5.6. Out-of-Sample Validation

We conduct a rolling-window out-of-sample evaluation using USDA NASS yield and USDA AMS price data from 2005 to 2022 across 15 Iowa counties. In each year, we estimate the copula model on the preceding five years of data and compute the CVaR-optimal allocation. We compare realized profits and realized shortfalls across four policies: CVaR-optimal, SU, EO, and MV. The CVaR-optimal policy achieves the lowest average shortfall in the worst-decile years (mean shortfall: \$18,600) relative to SU (\$29,200), EO (\$38,500), and MV (\$24,100). The average expected profit sacrifice relative to EO is \$6,800 per year, confirming that downside protection comes at a modest expected-profit cost. Ranking reversals (as defined in Definition 4.2) occur in 11 of the 18 county-year combinations with $\hat{\lambda}$ estimated above 0.58, consistent with the theory.

6. Discussion

The results of this paper have several concrete implications for the design of crop advisory systems, agricultural lending, and climate-adaptation policy. First, any recommendation system that outputs only a suitability ranking without an associated risk profile is potentially misleading for the subset of producers who operate under binding risk or liquidity constraints. A risk-aware recommendation system should augment the suitability score with three additional outputs: the marginal CVaR contribution of each crop at the recommended allocation, the lower-tail dependence coefficient with the existing portfolio, and the reversal threshold $\hat{\lambda}$ given the producer’s financial parameters. The computational cost of this augmentation is modest, since the CVaR optimization is a linear program.

Second, agricultural lenders who impose CVaR covenants on farm loans should recognize that tighter covenants lower the reversal threshold, making it more likely that the crop allocation supported by agronomic advice will violate the covenant. There is thus a structural tension between agronomic optimality and financial feasibility that lenders should anticipate in covenant design. Covenants calibrated to tail-dependent joint profit distributions will be less likely to generate infeasible agronomic recommendations.

Third, the information–flexibility complementarity result implies that policy programs aimed at improving climate forecasts for smallholder producers may have limited effectiveness if those producers face binding operational constraints from liquidity, land

tenure, or crop insurance availability. Complementary investments that relax operational constraints—credit access, flexible marketing contracts, equipment sharing programs—may be necessary prerequisites for climate information to generate its full social value.

The theory also suggests a novel evaluation criterion for crop recommendation system developers. Rather than benchmarking by predictive accuracy (how well the model forecasts yield), systems should be evaluated by their *decision quality*: how does following the recommendation affect the CVaR-adjusted profit of a constrained producer? This shift from predictive to prescriptive evaluation is consistent with the broader movement in operations research toward decision-focused machine learning (Elmachtoub and Grigas, 2022).

7. Conclusion

This paper develops a theory of ranking reversal in crop planning: the phenomenon whereby the agronomically most suitable crop is not the operationally optimal crop for a risk-constrained producer. We characterize the exact conditions under which reversal occurs and identify lower-tail dependence as the central mechanism. When extreme losses across crops co-move, concentrating on the highest-ranked crop amplifies portfolio tail risk in a way that a CVaR constraint penalizes more heavily than the expected-margin advantage can justify. This *diversification failure*—the breakdown of naive portfolio diversification under tail dependence—is formally distinct from the misspecification errors of mean-variance or expected-profit models, and we show it is systematically underestimated by Gaussian copula models that are otherwise observationally similar.

Our theory also establishes an information–flexibility complementarity: the operational value of better climate information is bounded above by the producer’s flexibility to act on it. For heavily constrained producers, precision agriculture investments may yield near-zero returns regardless of forecast quality. This result has direct implications for the targeting of technology subsidies and extension services.

Several extensions remain for future work. An equilibrium extension would examine whether homogeneous adoption of CVaR-optimal strategies by many producers induces endogenous price co-movement, potentially recreating the tail dependence that individual producers seek to avoid. A dynamic extension would examine multi-year acreage adjustment under evolving climate distributions. And an empirical structural-estimation

approach using farm-level data would allow direct estimation of the CVaR shadow price λ^* and reversal threshold $\hat{\lambda}$ for different producer types.

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